WTW 158 - Memo semester test 1 - 2013

SECTION A: MULTIPLE CHOICE QUESTIONS - 20 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 60 MINUTES.
2. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use SIDE 2 of the optic reader form. **(If side 1 is used it can not be marked.)**
   Fill in your personal information in pencil on side 2.
   Fill in your student number from top to bottom and then code it.
   If you make a mistake when coding your student number or if the student number is not filled in, you will get no marks for section A.
   Answer questions 1 to 20 on the optic reader form on **SIDE 2**.
6. **You may start with section B as soon as you have marked your answers on the optic reader form.**

AFDELING A : MEERVOUDIGE KEUSEVRAE - 20 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 60 MINUTE INHANDIG.
2. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie. Omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
   (Indien kant 1 gebruik word kan dit nie nagesien word nie).
   Vul u persoonlike inligting in potlood in op kant 2.
   Vul u studentenommer van bo na onder in en kodeer dit.
   Indien u studentenommer verkeerd gekodeer is of nie ingevul is nie, sal u nie punte vir afdeling A kry nie.
   Beantwoord vrae 1 tot 20 op die merkleesvorm se **KANT 2**.
6. **U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.**

Question 1 / Vraag 1

$|5x-6| \le 4 \Rightarrow -4 \le 5x-6 \le 4$

$\Rightarrow 2 \le 5x \le 10$

$\Rightarrow 2/5 \le x \le 2$

The solution set of $|5x-6| \le 4$ is

Die oplossingsversameling van $|5x-6| \le 4$ is

[1a] ϕ (empty set / leë versameling)	[1b] $\mathbb{R}$	[1c] $(-\infty, 2]$	[1d] $[\frac{2}{5}, \infty)$
[1e] $(-\infty, \frac{2}{5}] \cup [2, \infty)$	[1f] $[\frac{2}{5}, 2]$	[1g] None of these / Geen van hierdie	

Question 2 / Vraag 2

The solution set of $|x-1| > -1$ is

$|x-1| \geq 0$ for all x

Die oplossingsversameling van $|x-1| > -1$ is

[2a] ϕ (empty set / leë versameling)	[2b] $\mathbb{R}$	[2c] $(0,\infty)$	[2d] $(-\infty,2)$
[2e] $(-\infty,0)\cup(2,\infty)$	[2f] $(0,2)$	[2g] None of these / Geen van hierdie	

Question 3 / Vraag 3

$\Rightarrow |x-2| \neq 0$ and $x+5>0$

The solution set of $\dfrac{|x-2|}{x+5} > 0$ is

$\Rightarrow x \neq 2$ and $x > -5$

Die oplossingsversameling van $\dfrac{|x-2|}{x+5} > 0$ is

[3a] ϕ (empty set / leë versameling)	[3b] $\mathbb{R}$	[3c] $(-5,\infty)$	[3d] $(-\infty,-5)\cup(2,\infty)$
[3e] $(-5,2)\cup(2,\infty)$	[3f] $(-5,2)$	[3g] None of these / Geen van hierdie	

Question 4 / Vraag 4

The domain of the function $f(x) = \sqrt{2-x}$ is

$2-x \geq 0$

Die definisieversameling van die funksie $f(x) = \sqrt{2-x}$ is

$\Rightarrow x \leq 2$

[4a] $[0,\infty)$	[4b] $(2,\infty)$	[4c] $[2,\infty)$	[4d] $(-\infty,2)$
[4e] $(-\infty,2]$	[4f] None of these / Geen van hierdie		

Question 5 / Vraag 5

The range of the function $f(x) = (x-1)^{1/3}$ is

$\mathbb{R}$, shifted power function

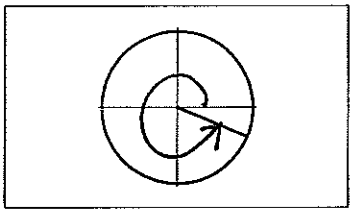
Die waardeversameling van die funksie $f(x) = (x-1)^{1/3}$ is

[5a] $[0,\infty)$	[5b] $[1,\infty)$	[5c] $(-\infty,1]$	[5d] $(-\infty,\infty)$
[5e] None of these / Geen van hierdie			

Question 6 / Vraag 6

The angle in the sketch is

Die hoek in die skets is

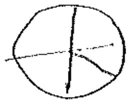


[6a] $\frac{\pi}{8}$	[6b] $\frac{7\pi}{8}$	[6c] $\frac{9\pi}{8}$	[6d] $\frac{15\pi}{8}$
[6e] $-\frac{\pi}{8}$	[6f] $-\frac{7\pi}{8}$	[6g] $-\frac{9\pi}{8}$	[6h] $-\frac{15\pi}{8}$

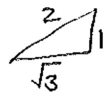
Question 7 / Vraag 7

$\tan(-\frac{\pi}{6}) =$

[7a] 1	[7b] -1	[7c] $\sqrt{3}$	[7d] $-\sqrt{3}$	[7e] $\frac{1}{\sqrt{3}}$	[7f] $-\frac{1}{\sqrt{3}}$
[7g] None of these / Geen van hierdie					



tan neg



Question 8 / Vraag 8

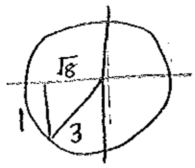
$\sin(\frac{13\pi}{2}) =$ $\sin\left(\frac{13\pi}{2} - 3\times 2\pi\right) = \sin\frac{\pi}{2} = 1$

[8a] 1	[8b] -1	[8c] 0	[8d] None of these / Geen van hierdie

Question 9 / Vraag 9

If $\sin x = -\frac{1}{3}$, $\pi < x < \frac{3\pi}{2}$, then $\cot x =$

As $\sin x = -\frac{1}{3}$, $\pi < x < \frac{3\pi}{2}$, then $\cot x =$



$\cot x = \frac{1}{\tan x}$

$= \frac{1}{\frac{1}{\sqrt{8}}}$

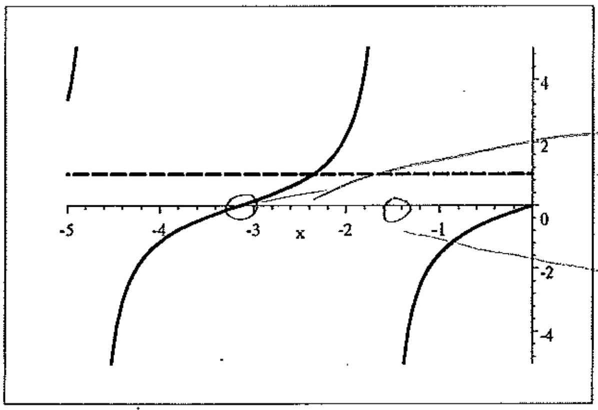
$= \sqrt{8}$

[9a] $\sqrt{8}$	[9b] $-\sqrt{8}$	[9c] $\frac{1}{\sqrt{8}}$	[9d] $\frac{-1}{\sqrt{8}}$	[9e] $\frac{3}{\sqrt{8}}$	[9f] $\frac{-3}{\sqrt{8}}$
[9g] None of these / Geen van hierdie					

Question 10 / Vraag 10

The graphs of $y = \tan x$ and $y = 1$ below are sketched on the intervals $x \in [-5, 0]$ and $y \in [-5, 5]$. The graphs intersect at the point where $x =$
Hint: $\tan x = 1 \Rightarrow x = \frac{\pi}{4} + k\pi, k \in \mathbb{Z}$

Die grafieke van $y = \tan x$ en $y = 1$ hieronder is op die intervalle $x \in [-5, 0]$ en $y \in [-5, 5]$ geskets. Die grafieke sny in die punt waar $x =$
Wenk: $\tan x = 1 \Rightarrow x = \frac{\pi}{4} + k\pi, k \in \mathbb{Z}$



from graph.

$\to \tan(-\pi) = 0$

$\tan(-\frac{\pi}{2})$ is undefined

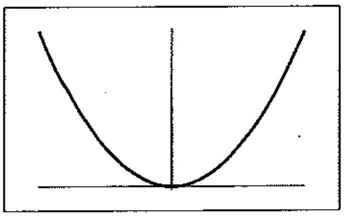
$\therefore\ x \in (-\pi, -\frac{\pi}{2})$

[10a] $\frac{\pi}{4}$	[10b] $\frac{5\pi}{4}$	[10c] $-\frac{7\pi}{4}$	[10d] $-\frac{3\pi}{4}$
[10e] None of these / Geen van hierdie			

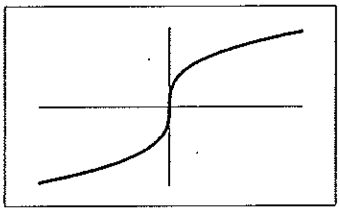
Question 11 / Vraag 11

Consider the graphs of the functions below. The graph of $y = x^5$ is number

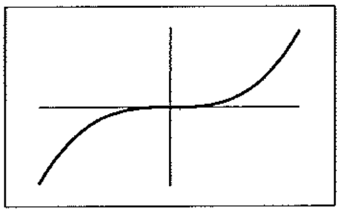
Beskou die grafieke van die funksies hieronder. Die grafiek van $y = x^5$ is nommer



I



II



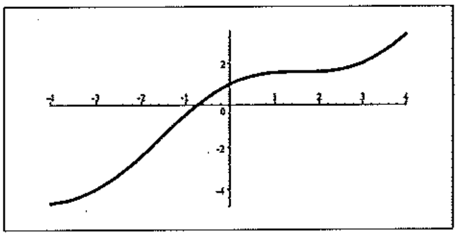
III

[11a] I	[11b] II	[11c] III	[11d] None of these / Geen van hierdie

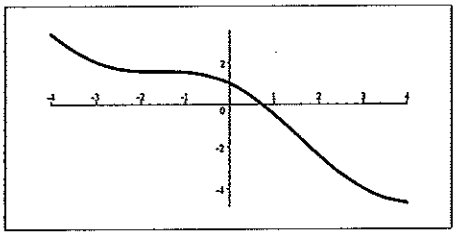
Question 12 / Vraag 12

Consider the graphs below. The graph of $y = -f(x)$ is

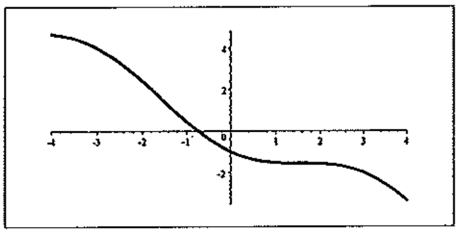
Beskou die grafieke hieronder. Die grafiek van $y = -f(x)$ is



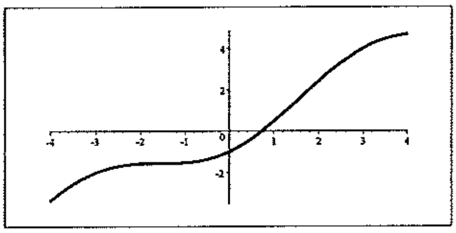
$y = f(x)$



$y = g_1(x)$



$y = g_2(x)$

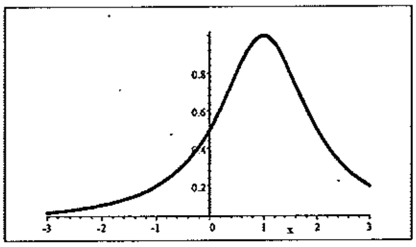


$y = g_3(x)$

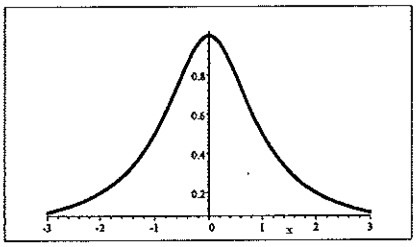
[12a] $y = g_1(x)$	[12b] $y = g_2(x)$	[12c] $y = g_3(x)$

Question 13 / Vraag 13

Consider the graphs below. / Beskou die grafieke hieronder.



$y = f(x)$



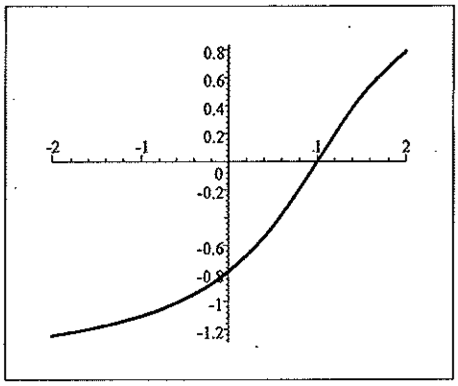
$y = g(x)$

$g(x) =$

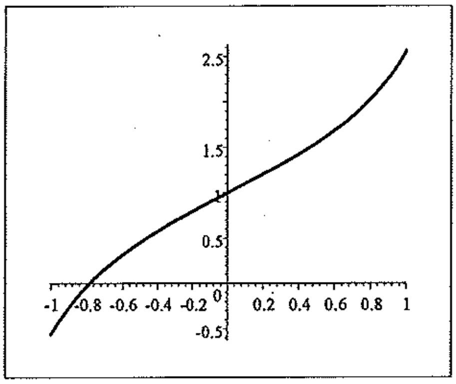
[13a] $f(x-1)$	[13b] $f(x+1)$	[13c] $f(x) - 1$	[13d] $f(x) + 1$

Question 14 / Vraag 14

Consider the graphs below. / Beskou die grafieke hieronder..



$y = f(x)$



$y = g(x)$

$f(1)=0$
$\Rightarrow f^{-1}(0)=1$
and $f(0)=-0.8$
$\Rightarrow f^{-1}(-0.8)=0$

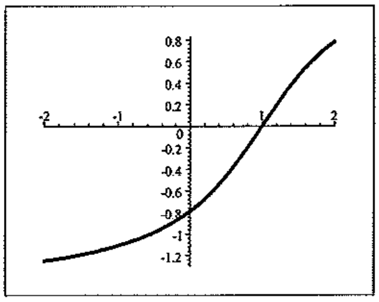
$g(x) =$

[14a] $f(x-1.8)$	[14b] $f(x+1.8)$	[14c] $f^{-1}(x)$	[14d] $f(x)+1.8$

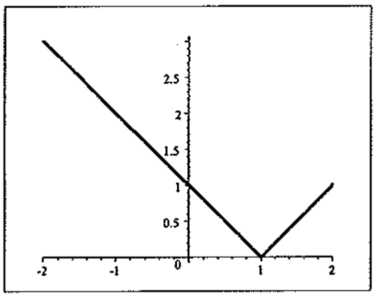
Question 15 / Vraag 15

Consider the graphs below. Which graph is the graph of $y = |f(x)|$?

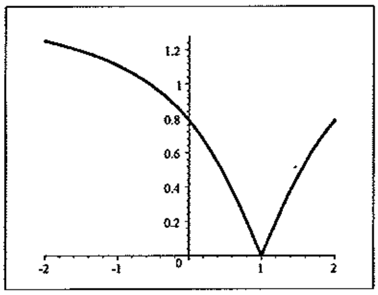
Beskou die grafieke hieronder. Watter grafiek is die grafiek van $|f(x)|$?



$y = f(x)$



$y = g(x)$



$y = h(x)$

[15a] $g(x)$	[15b] $h(x)$	[15c] None of these / Geen van hierdie

Question 16 / Vraag 16

If $(f \circ g \circ h)(x) = \sin^2 \sqrt{x}$, then / As $(f \circ g \circ h)(x) = \sin^2 \sqrt{x}$, dan is

[16a]	$f(x) = \sqrt{x}$, $g(x) = \sin x$ and / en $h(x) = x^2$
[16b]	$f(x) = \sqrt{x}$, $g(x) = \sin\sqrt{x}$ and / en $h(x) = \sin^2\sqrt{x}$
[16c]	$f(x) = x^2$, $g(x) = \sin x$ and / en $h(x) = \sqrt{x}$
[16d]	$f(x) = x^2$, $g(x) = \sqrt{x}$ and / en $h(x) = \sin x$
[16e]	$f(x) = \sin x$, $g(x) = x^2$ and / en $h(x) = \sqrt{x}$
[16f]	$f(x) = \sin x$, $g(x) = \sqrt{x}$ and / en $h(x) = x^2$
[16g]	None of these / Geen van hierdie

Question 17 / Vraag 17

$2(3x)+2x-5x$
$=3x$

If $\dfrac{(e^{3x})^2 e^{2x}}{e^{5x}} = e^a$, then $a =$ / As $\dfrac{(e^{3x})^2 e^{2x}}{e^{5x}} = e^a$, dan is $a =$

[17a] $3x$	[17b] $9x^2-3x$	[17c] $\frac{18}{5}x^2$	[17d] None of these / Geen van hierdie

Question 18 / Vraag 18 $= \ln((x^3)^2) + \ln 5x - \ln x^4 = \ln \frac{x^6 5x}{x^4}$

If $2\ln x^3 - 4\ln x + \ln 5x = \ln a$, then $a =$ / As $2\ln x^3 - 4\ln x + \ln 5x = \ln a$, dan is $a =$

[18a] $2x^3+x$	[18b] $5x^3$	[18c] $\frac{x}{5}$	[18d] None of these / Geen van hierdie

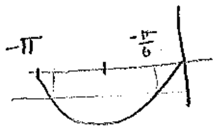
Question 19 / Vraag 19

The solution set of $2\sin x + 1 < 0,\ x \in [-\pi, 0]$ is

Die oplossingsversameling van $2\sin x + 1 < 0,\ x \in [-\pi, 0]$ is

$\sin x < -\frac{1}{2}$
$\sin x = -\frac{1}{2}$
$\Rightarrow x = \frac{7\pi}{6} + 2n\pi$ or $x = -\frac{\pi}{6} + 2n\pi$

[19a] $(\frac{7\pi}{6}, \frac{11\pi}{6})$	[19b] $(\frac{4\pi}{3}, \frac{5\pi}{3})$	[19c] $(\frac{-5\pi}{6}, \frac{-\pi}{6})$
[19d] $(\frac{-2\pi}{3}, \frac{-\pi}{3})$	[19e] None of these / Geen van hierdie	



$\frac{7\pi}{6} - 2\pi$
$= -\frac{5\pi}{6}$

Question 20 / Vraag 20

$\ln(x^2) > 6 \Rightarrow x^2 > e^6$
$\Rightarrow x^2 - e^6 > 0$

The solution set of $\ln(x^2) - 6 > 0$ is

Die oplossingsversameling van $\ln(x^2) - 6 > 0$ is

[20a] (e^3, ∞)	[20b] $(-\infty, e^3)$	[20c] $(-\infty, -e^3)$	[20d] $(-\infty, -e^3) \cup (e^3, \infty)$
[20e] $(-e^3, e^3)$	[20f] None of these / Geen van hierdie		

[20]

SECTION B: 30 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following questions on this paper **in ink.**
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked. If you need more space for an answer, use the facing page and indicate it clearly **by framing it.**
5. Show all steps and calculations and explain your answers.

AFDELING B: 30 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel **in ink.**
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie. As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek.**
5. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 21 / Vraag 21

Solve for x if $\frac{x-2}{3-x} \leq 0$.

You may not use a table or graphs to solve the inequality.

Write your answer in interval notation.

Los op vir x as $\frac{x-2}{3-x} \leq 0$.

U mag nie 'n tabel of grafieke gebruik om die ongelykheid op te los nie.

Gee die antwoord in intervalnotasie.

$\frac{x-2}{3-x} \leq 0$

$\Rightarrow x-2 \leq 0$ and $3-x > 0$ or $x-2 \geq 0$ and $3-x < 0$

$\Rightarrow x \leq 2$ and $x < 3$ or $x \geq 2$ and $x > 3$

$\Rightarrow x \leq 2$ (2, 3) or $x > 3$ (2, 3)

$\Rightarrow x \in (-\infty, 2] \cup (3, \infty)$

[3]

Question 22 / Vraag 22

Solve for x if $2\cos x \sin x = -\cos x$.

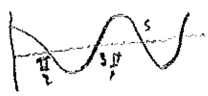
Los op vir x as $2\cos x \sin x = -\cos x$.

$2\cos x \sin x = -\cos x \Rightarrow 2\cos x \sin x + \cos x = 0$

$\Rightarrow \cos x(2\sin x + 1) = 0$

$\Rightarrow \cos x = 0$ or $2\sin x + 1 = 0$

$\cos x = 0 \Rightarrow x = \frac{\pi}{2} + n\pi = (2n+1)\frac{\pi}{2}$



$n \in \mathbb{Z}$

[4]

$\sin x = -\frac{1}{2}$; Reference Angle $= \frac{\pi}{6}$

$\sin x = -\frac{1}{2} \Rightarrow x = (\pi + \frac{\pi}{6}) + 2n\pi = \frac{7\pi}{6} + 2n\pi$

or $x = (2\pi - \frac{\pi}{6}) + 2n\pi = \frac{11\pi}{6} + 2n\pi$

$\therefore$ Solutions: $x = \frac{\pi}{2} + n\pi$ or $x = \frac{7\pi}{6} + 2n\pi$

or $x = \frac{11\pi}{6} + 2n\pi$ $n \in \mathbb{Z}$

Question 23 / Vraag 23

i Solve for x if $\ln x + \ln(x-1) = \ln 2$.

Los op vir x indien $\ln x + \ln(x-1) = \ln 2$.

$\ln x + \ln(x-1) = \ln 2$ Domain $= \{x \mid x>0 \text{ and } x-1>0\}$ $= (1,\infty)$

$\Rightarrow \ln x(x-1) = \ln 2$ property of ln

$\Rightarrow e^{\ln(x^2-x)} = e^{\ln 2}$

$\Rightarrow x^2 - x = 2$ cancellation law

$\Rightarrow x^2 - x - 2 = (x-2)(x+1) = 0$

$\Rightarrow x = 2$ or $x = -1$

BUT -1 is not in the domain $\therefore$ $x = 2$

[4]

ii Find a function of the form $f(x) = ae^{bx}$ with $f(0) = 2$ and $f(3) = 6$. Show all the calculations.

Bepaal 'n funksie van die vorm $f(x) = ae^{bx}$ met $f(0) = 2$ en $f(3) = 6$. Toon al die berekeninge.

$f(0) = 2 \Rightarrow ae^0 = 2 \Rightarrow \underline{a = 2}$

$f(3) = 6 \Rightarrow ae^{3b} = 6$

$\Rightarrow 2e^{3b} = 6$

$\Rightarrow e^{3b} = 3$

$\Rightarrow \ln e^{3b} = \ln 3$

$\Rightarrow 3b = \ln 3$

$\Rightarrow b = \frac{1}{3}\ln 3$

$\therefore f(x) = 2e^{(\frac{1}{3}\ln 3)x}$

[4]

Question 24 / Vraag 24

i Write down the definition of the absolute value of a real number x.

Gee die definisie van die absolute waarde van 'n reële getal x.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

[1]

ii Write / Skryf $\sqrt{|x-1|+4}$ in the form / in die vorm $\begin{cases} \ldots & \text{if / as } x \geq \ldots \\ \ldots & \text{if / as } x < \ldots \end{cases}$

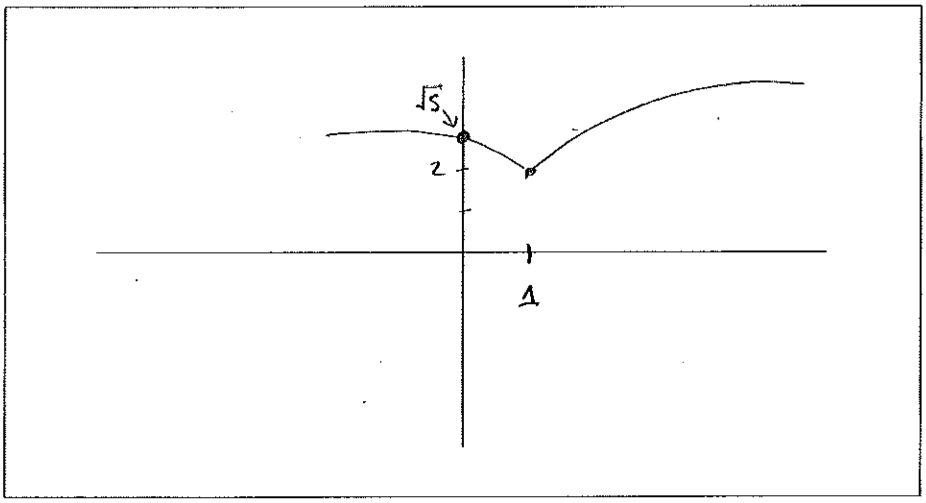
$$\sqrt{|x-1|+4} = \begin{cases} \sqrt{(x-1)+4} & \text{if } x-1 \geq 0 \\ \sqrt{-(x-1)+4} & \text{if } x-1 < 0 \end{cases}$$

$$= \begin{cases} \sqrt{x+3} & \text{if } x \geq 1 \\ \sqrt{-x+5} & \text{if } x < 1 \end{cases}$$

[2]

iii Sketch the graph of the function $f(x) = \sqrt{|x-1|+4}$. Indicate the intercepts with the axes.

Skets die grafiek van die funksie $\sqrt{|x-1|+4}$. Dui die snypunte met die asse aan.



[3]

iv Write down the domain and range of the function in iii above.

Gee die definisieversameling en waardeversameling van die funksie in iii hierbo.

Domain / Definisieversameling = $\mathbb{R} = (-\infty, \infty)$

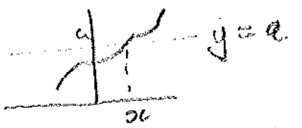
Range / Waardeversameling = $[2, \infty) = \{y \mid y \geq 2\}$

[2]

Question 25 / Vraag 25

Let f be a one-to-one function such that the graph of f intersects every line $y = a,\ a \neq 3$.
Write down the domain of f^{-1} and explain your answer.

Laat f 'n een-tot-een funksie wees sodanig dat die grafiek van f elke lyn $y = a,\ a \neq 3$, sny.
Gee die definisieversameling van f^{-1} en verduidelik u antwoord.



If the graph of f intersects the line $y=a$ there is a x in the domain of f such that $f(x)=a$

$\therefore$ a is in the range of f

$\therefore$ Domain of f^{-1} = range of f = $\{a \mid a \neq 3\}$

[3]

Question 26 / Vraag 26

i Write down the definition of an odd function.

Gee die definisie van 'n onewe funksie

f is an odd function if

$f(-x) = -f(x)$ for every x in the domain.

[1]

ii Assume that f is an odd function with domain $\mathbb{R}$.
Is $f \circ f$ also an odd function? Show all the calculations and explain your answer.

Aanvaar dat f 'n onewe funksie is met definisieversameling $\mathbb{R}$.
Is $f \circ f$ ook 'n onewe funksie? Toon alle berekeninge en verduidelik u antwoord.

[3]

$(f \circ f)(-x) = f(f(-x))$

$= f(-f(x))$ f is odd

$= -f(f(x))$ f is odd

$= -(f \circ f)(x)$

$\therefore$ Yes $f \circ f$ is odd.